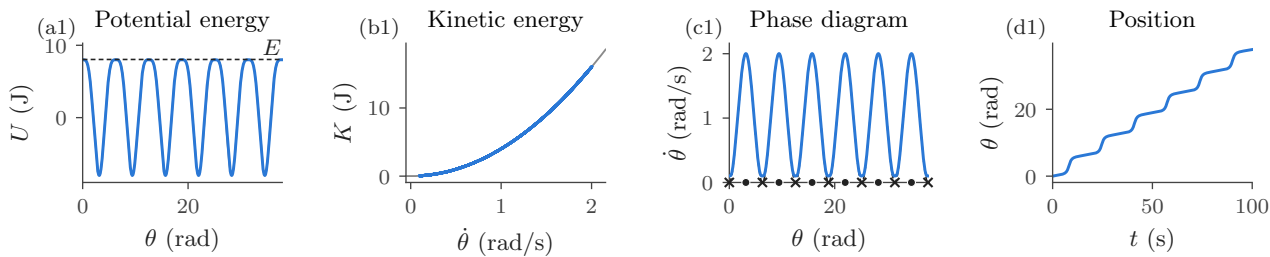
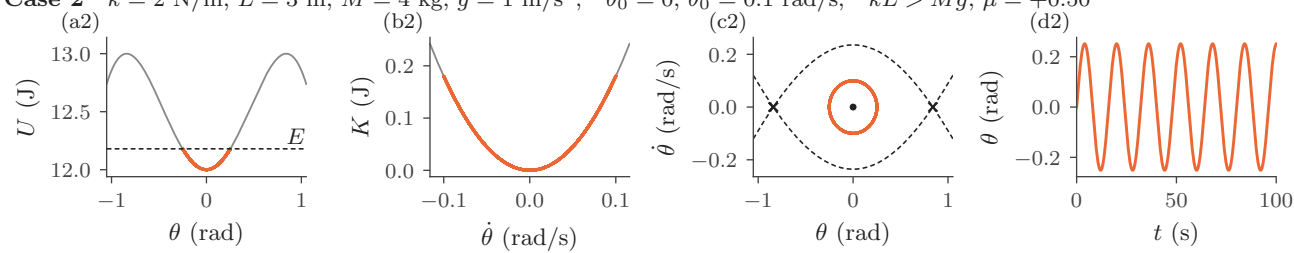


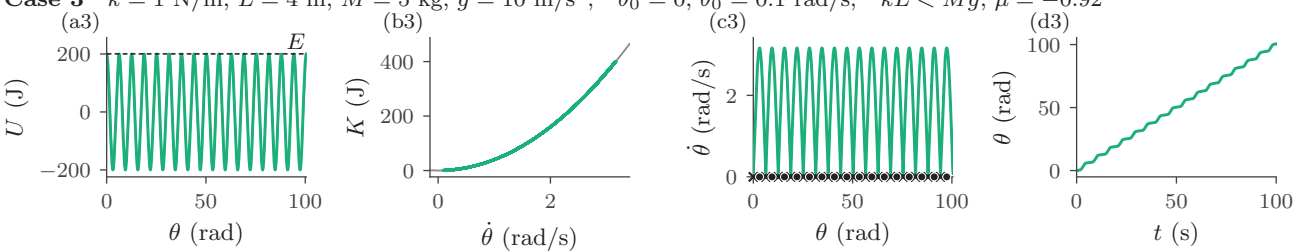
Case 1 $k = 2 \text{ N/m}$, $L = 2 \text{ m}$, $M = 2 \text{ kg}$, $g = 2 \text{ m/s}^2$; $\theta_0 = 0$, $\dot{\theta}_0 = 0.1 \text{ rad/s}$; $kL = Mg$, $\mu = 0$



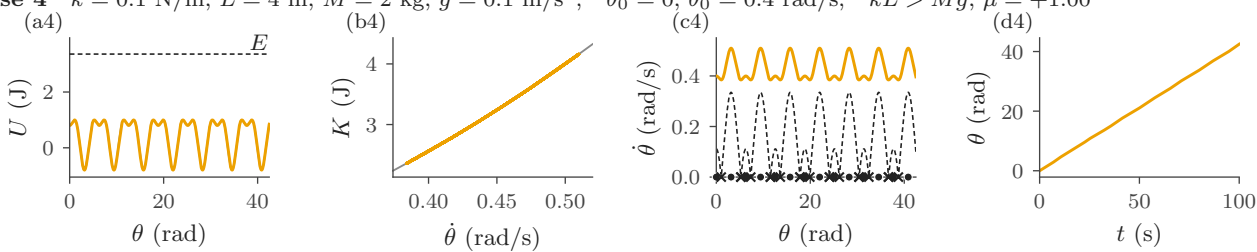
Case 2 $k = 2 \text{ N/m}$, $L = 3 \text{ m}$, $M = 4 \text{ kg}$, $g = 1 \text{ m/s}^2$; $\theta_0 = 0$, $\dot{\theta}_0 = 0.1 \text{ rad/s}$; $kL > Mg$, $\mu = +0.50$



Case 3 $k = 1 \text{ N/m}$, $L = 4 \text{ m}$, $M = 5 \text{ kg}$, $g = 10 \text{ m/s}^2$; $\theta_0 = 0$, $\dot{\theta}_0 = 0.1 \text{ rad/s}$; $kL < Mg$, $\mu = -0.92$



Case 4 $k = 0.1 \text{ N/m}$, $L = 4 \text{ m}$, $M = 2 \text{ kg}$, $g = 0.1 \text{ m/s}^2$; $\theta_0 = 0$, $\dot{\theta}_0 = 0.4 \text{ rad/s}$; $kL > Mg$, $\mu = +1.00$



Case 5 $k = 0.1 \text{ N/m}$, $L = 3 \text{ m}$, $M = 5 \text{ kg}$, $g = 0.1 \text{ m/s}^2$; $\theta_0 = 0$, $\dot{\theta}_0 = 0.4 \text{ rad/s}$; $kL < Mg$, $\mu = -0.40$

